PX-400 Field Service Manual

Fault E-217 Troubleshooting and Diagnostic D-17

Oriole Packaging Systems — Service Engineering

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1 Safety

Isolate mains power and lock out the line before touching the encoder connector, harness, or controller cabinet. Verify zero energy with a meter. Never reseat connectors with power applied. Record every intervention in the service work order before leaving the site.

2 Fault code reference

Code	Meaning	First action
E-210	Index overtravel	Check mechanical stops, then home offset
E-217	Encoder signal loss during index	See Section 3, do not replace parts first
E-223	Controller watchdog timeout	Capture log, cycle control power
E-301	Firmware image mismatch	Verify firmware version, reflash per release note

3 E-217 troubleshooting procedure

E-217 means the motion controller lost a valid encoder signal during an indexed movement. After firmware 4.8 the diagnostic sampling window is tighter, so marginal encoder connections that used to recover now raise a stop. Work the steps in order and stop at the first step that resolves the fault.

1. Read the controller fault log and note the E-217 timestamps. Compare them with recent firmware or maintenance changes.
2. Isolate power. Inspect the encoder connector (J7), harness routing, shield termination, and visible cable damage along the full cable run.
3. Reseat the J7 connector. Check the locking tab and confirm the shield drain wire is terminated at the controller end only.
4. Re-apply power and run Diagnostic D-17 (Section 4). Record pass or fail with the fault log attached to the work order.
5. If D-17 fails, replace the indicated component per Section 5 and rerun D-17.
6. If D-17 passes repeatedly but E-217 stops continue, capture the controller log and escalate to engineering with timestamps and the D-17 results.

Do not downgrade firmware as the first response to E-217.

4 Diagnostic D-17 procedure

D-17 exercises the index axis through ten slow index cycles while sampling the encoder channel each millisecond. A pass requires ten clean cycles with no signal dropout longer than 2 ms. A fail reports the cycle number and the dropout duration; attach this output to the work order as the diagnostic basis for any part replacement.

5 Replacement parts

SKU	Description	Replace when
EA-400	PX-400 encoder assembly	D-17 fails after connector/harness check
HN-44	Shielded encoder harness	Visible damage or shield fault found
PS-12	Controller power module	Only on power-rail fault, never for E-217 alone

6 Connector and torque reference

Item	Specification
Encoder connector	J7, 17-pin circular, locking tab
J7 coupling torque	1.2 N m
Cabinet ground lug	2.5 N m
Shield termination	Controller end only, drain wire under clamp